SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse realworld problem.*		
Student Name:	K.Kabilan	**Reg. No.:**	192521157
Section / Batch:	B.tech-IT	**Date:**	31/08/2026

Code of Conduct

I _________K.Kabilan (192521157)_______________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________K.Kabilan___________

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20

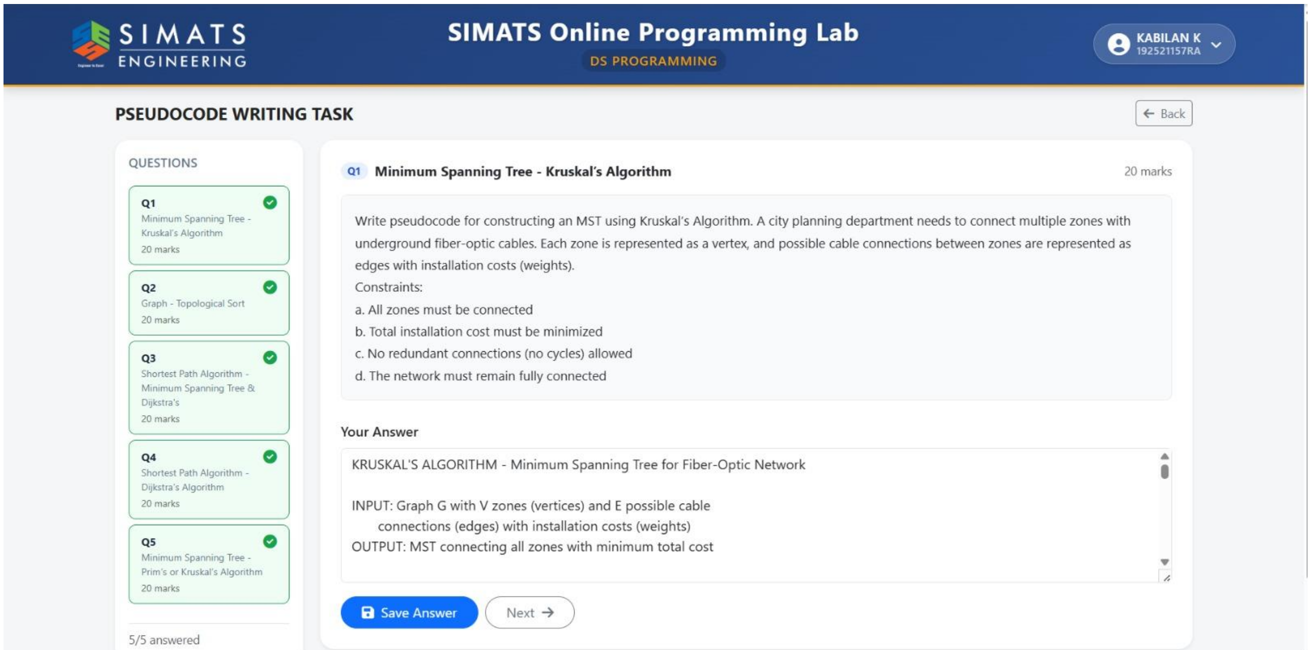
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
	GRAND TOTAL:	**100 Marks**	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or nonfunctional code
Code Quality & Readability	20	Clean, modular, wellcommented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

1)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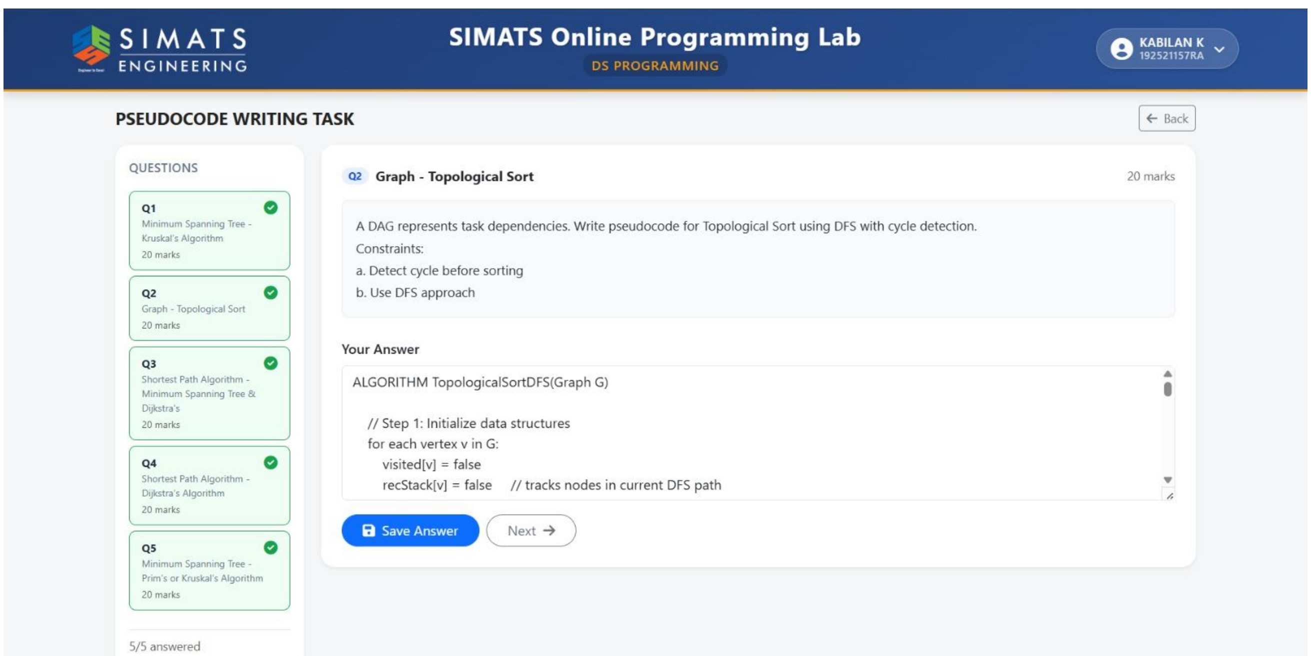
QUESTIONS
Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks
Q2
Graph - Topological Sort
20 marks
Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks
Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
5/5 answered
Q1 Minimum Spanning Tree - Kruskal's Algorithm
20 marks
Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).
Constraints:
a. All zones must be connected
b. Total installation cost must be minimized
c. No redundant connections (no cycles) allowed
d. The network must remain fully connected
Your Answer
KRUSKAL'S ALGORITHM - Minimum Spanning Tree for Fiber-Optic Network
INPUT: Graph G with V zones (vertices) and E possible cable
connections (edges) with installation costs (weights)
OUTPUT: MST connecting all zones with minimum total cost
Save Answer
Next

2)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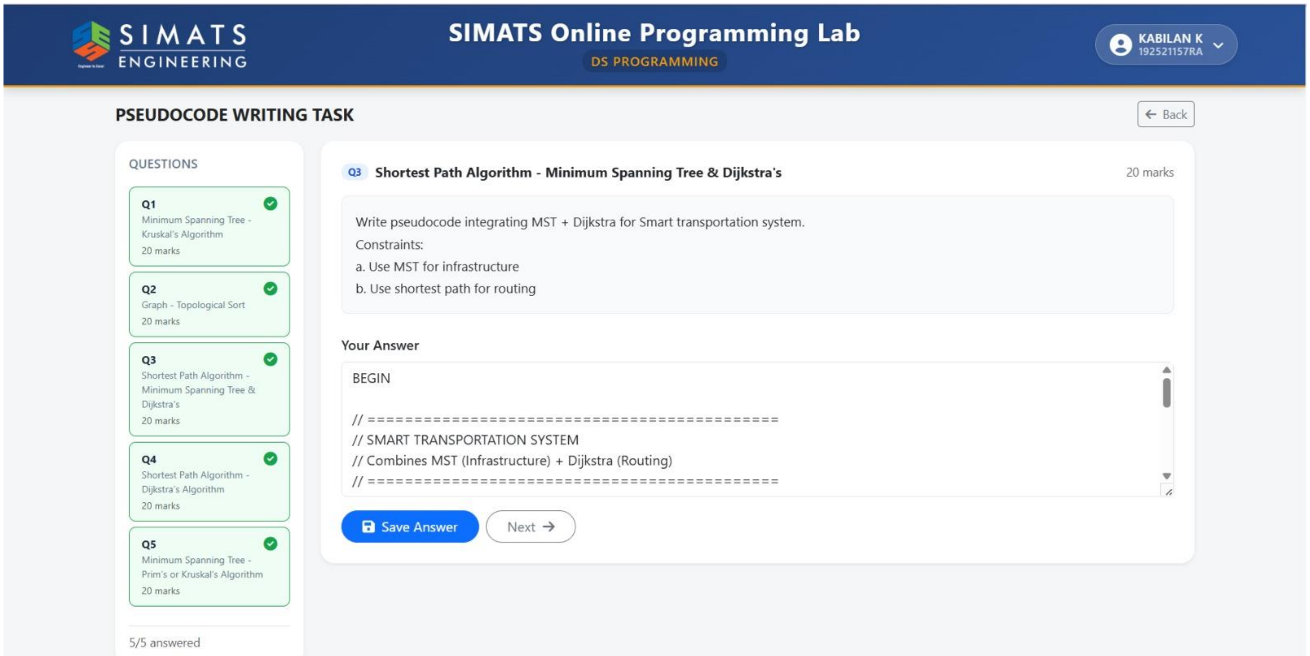
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
5/5 answered
Q2 Graph - Topological Sort
20 marks
A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.
Constraints:
a. Detect cycle before sorting
b. Use DFS approach
Your Answer
ALGORITHM TopologicalSortDFS(Graph G)
// Step 1: Initialize data structures
for each vertex v in G:
visited[v] = false
recStack[v] = false // tracks nodes in current DFS path
Save Answer
Next

3)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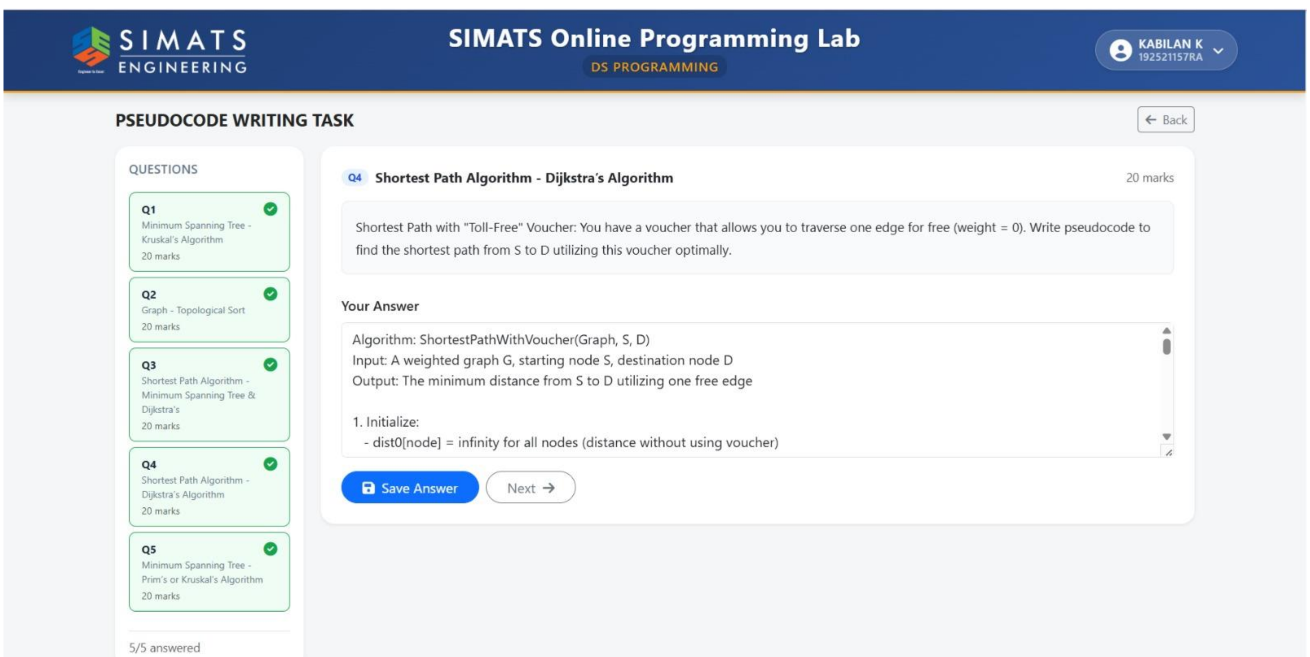
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
5/5 answered
Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks
Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing
Your Answer
BEGIN
// ==
// SMART TRANSPORTATION SYSTEM
// Combines MST (Infrastructure) + Dijkstra (Routing)
// ==
Save Answer
Next

4)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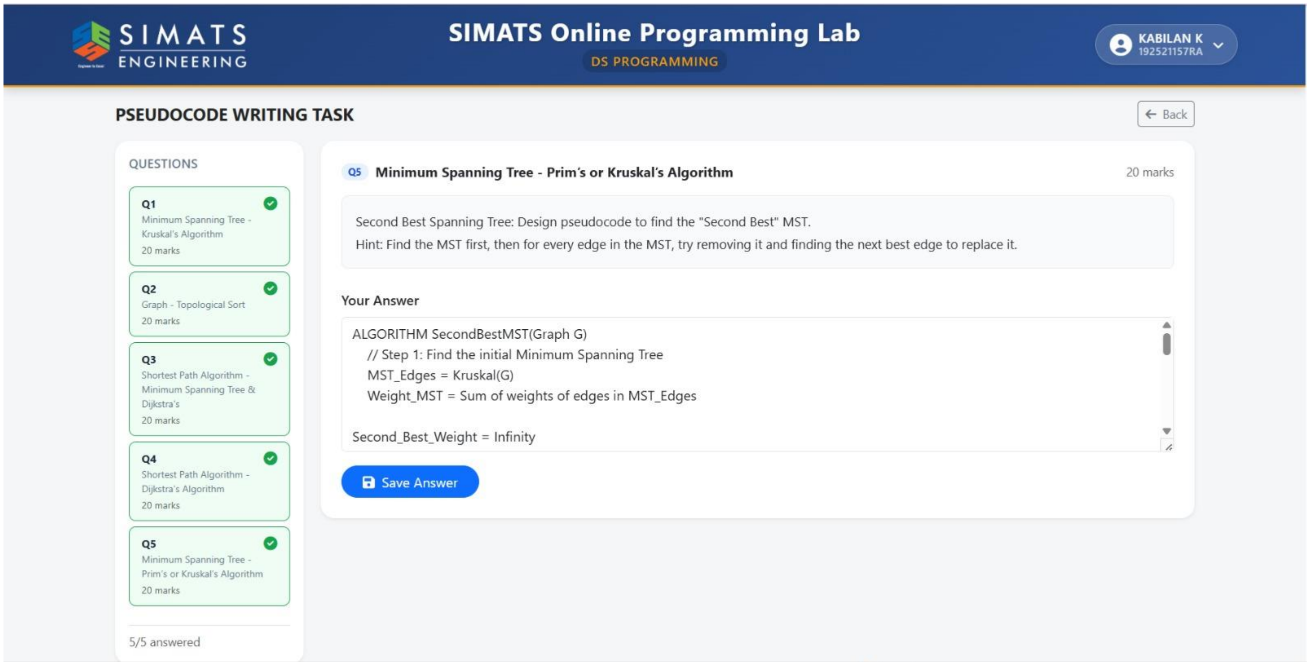
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
5/5 answered
Q4 Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.
Your Answer
Algorithm: ShortestPathWithVoucher(Graph, S, D)
Input: A weighted graph G, starting node S, destination node D
Output: The minimum distance from S to D utilizing one free edge
1. Initialize:
- dist0[node] = infinity for all nodes (distance without using voucher)
Save Answer
Next

5)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Shortest Path Algorithm - Dijkstra's Algorithm
20 marks
Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
5/5 answered
Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks
Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.
Your Answer
ALGORITHM SecondBestMST(Graph G)
// Step 1: Find the initial Minimum Spanning Tree
MST_Edges = Kruskal(G)
Weight_MST = Sum of weights of edges in MST_Edges
Second_Best_Weight = Infinity
Save Answer